Prakhar **Verma**

Machine Learning Researcher · PhD

Helsinki, Finland.

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Research Summary

Machine learning researcher in probabilistic modelling, Bayesian inference, and large language models, with publications at ICML (Oral), NeurIPS, AISTATS, and TMLR. Building on prior research experience at Microsoft Research, Adobe Research, and the University of Oxford, I currently build reliable, production-scale LLM systems at Inven, developing theoretically grounded methods with practical impact. I am particularly interested in the intersection of probabilistic modelling and agentic systems, with the goal of building principled, uncertainty-aware agent frameworks for evaluation, failure detection and recovery, and decision-making under uncertainty.

Experience

Inven
Helsinki, Finland

Senior Machine Learning Engineer — *October 2025 - Present*

- Researched and developed a document-processing pipeline that converts millions of private, multimodal documents into a hierarchically structured corpus, enabling **agentic information retrieval**.
- Designed a **Bayesian entity-matching model** that links private companies to official records, fusing multiple noisy signals from heterogeneous, incomplete data sources.
- Developed a statistical model to estimate **company revenue** at scale by identifying global peer groups and inferring estimates conditioned on comparable firms.

Adobe Research
Bangalore, India

Research Intern — *June 2024 - August 2024*

- Developed a **Bayesian framework for sequential causal discovery** that integrates language model priors with observational data for uncertainty-aware structure learning.
- Proposed an iterative causal refinement approach to mitigate **dual-bias** arising from both language model priors and observational data.
- Led to a patent-pending method and a publication at TMLR.

Microsoft Research
Bangalore, India

Research Intern — *March 2024 - May 2024*

- Developed **Plan*RAG**, a framework for multi-hop **retrieval-augmented generation (RAG)** that builds **traceable reasoning DAGs** for efficient retrieval and inference.
- Reduced inference latency and retrieval cost through **efficient test-time planning** strategies.
- Led to a publication at the ICLR 2025 Workshop on Reasoning and Planning for LLMs.

University of Oxford
Oxford, United Kingdom

Visiting Researcher — *July 2023 - September 2023*

- Collaborated with Prof. Seth Flaxman and Elizaveta Semenova on **encoding prior knowledge** into probabilistic models for life science applications.
- Developed **scalable inference techniques** for Bayesian generative models in epidemiology and medical applications.
- Led to a research publication on scalable Bayesian deep generative models.

SpectacularAI
Espoo, Finland

Research Engineer — *September 2021 - September 2022*

- Developed **uncertainty estimation** methods for deep learning models using approximate Bayesian inference techniques, including MC Dropout and Laplace approximation, to improve model robustness under uncertainty.

TomTom
Pune, India

Software Engineer (R&D) — *July 2016 - August 2019*

- Developed a **semantic segmentation** pipeline for extracting map features from satellite imagery and integrating them into large-scale mapping databases.

Selected Publications

- **Prakhar Verma**, David Arbour, Sunav Choudhary, Harshita Chopra, Arno Solin, Atanu R. Sinha. Sequential Causal Discovery with Noisy Language Model Priors. *Transactions on Machine Learning Research (TMLR), 2026.*
- **Prakhar Verma**, Sukruta Prakash Midigeshi, Gaurav Sinha, Arno Solin, Nagarajan Natarajan, Amit Sharma. Plan*RAG: Efficient Test-Time Planning for Retrieval Augmented Generation. Workshop on *Reasoning and Planning for Large Language Models* at ICLR 2025.
- **Prakhar Verma**, Vincent Adam, Arno Solin. Variational Gaussian Process Diffusion Processes. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.*
- Paul Edmund Chang*, **Prakhar Verma***, S.T. John, Arno Solin, Mohammad Emtiyaz Khan. Memory-based dual Gaussian processes for sequential learning. *International Conference on Machine Learning (ICML), 2023. (Oral Presentation)*
- Arno Solin, Ella Tamir, **Prakhar Verma**. Scalable Inference in SDEs by Direct Matching of the Fokker–Planck–Kolmogorov Equation. *Advances in Neural Information Processing Systems 35* (NeurIPS), 2021.
- **Prakhar Verma**, Paul Chang, Arno Solin, Mohammad Emtiyaz Khan. Sequential Learning in GPs with Memory and Bayesian Leverage Score. *Asian Conference in Machine Learning (ACML) workshop "Continual Lifelong Learning" 2022 (Contributed talk)*.
- Elizaveta Semenova, **Prakhar Verma**, Max Cairney-Leeming, Arno Solin, Samir Bhatt, Seth Flaxman. PriorCVAE: Scalable MCMC parameter inference with Bayesian deep generative modelling.

Education

Aalto University
Finland

DOCTOR OF PHILOSOPHY (PH.D.) *2022 - 2026*

- Research on **probabilistic machine learning**, **Bayesian inference**, and **uncertainty quantification** with applications to sequential decision-making and large language models (LLMs). Supervised by Prof. Arno Solin; defended August 2026.
- **Thesis:** Scalable Probabilistic Inference for Sequential Stochastic Models

Aalto University
Finland

MASTER OF SCIENCE (M.SC.) - 4.7/5 (PASS WITH HONORS) *2019 - 2021*

- Major in Machine Learning, Data Science, and Artificial Intelligence and Minor in Mathematics
- **Thesis:** *Sparse Gaussian Processes for Stochastic Differential Equations*

Uttarakhand Technical University
India

BACHELOR OF TECHNOLOGY (B.TECH.) - 79.03% (FIRST DIVISION WITH HONORS) *2012 - 2016*

- Specialization in Information Technology
- **Thesis:** *Development of Automated GIS Tools on Various Platforms* with TomTom, India.

Research Interests & Technical Skills

- **Research Interests:** Probabilistic Machine Learning, Bayesian Inference, Large Language Models, Retrieval-Augmented Generation, Sequential Decision-Making, Generative Modeling, Causal Discovery, Agentic Systems, Uncertainty Quantification
- **Frameworks & Tools:** PyTorch, JAX, Hugging Face Transformers, TensorFlow, GPflow, PySpark, Docker, AWS, Weights & Biases, Hydra
- **Programming Languages:** Python, Java, C

Academic Service

- Workflow Chair, International Conference on Artificial Intelligence and Statistics (AISTATS), 2026
- Regular reviewer for NeurIPS, AISTATS, ICLR, ICML, CoLLAs

Selected Honors

- Awarded *"Nokia Scholarship"* in *2024* for exceptional progress and research excellence during doctoral studies.
- Awarded *"Dean Scholarship"* in *2020* and *2021* at Aalto University for commendable academic progress during MSc studies.
- Winner of TomTom *"Innovation Day 2018"*, presented an AI plugin which bridges the gap between machines and cartographers.